# Darius Muglich

Machine learning researcher · DPhil student, University of Oxford

dariusmuglich.com / Google Scholar

Research in reinforcement learning, multi-agent systems, generalization, and optimization.

Programming & frameworks: Python, C++, PyTorch, JAX.

## Education

**University of Oxford** 2023–present · Expected September 2027

DPhil · Autonomous Intelligent Machines and Systems (AIMS CDT)

Supervisor: Jakob Foerster

Thesis on policy selection in reinforcement learning: how learning biases and structural constraints shape learned behaviour and generalization.

Current work on logic gate networks and neurosymbolic robot control.

**University of Oxford** 2020–2021

MSc · Advanced Computer Science

**University of British Columbia** 2015–2020

BSc · Combined Honours in Mathematics and Computer Science

## Selected publications

**Ad-Hoc Teamwork from Human Demonstrations** NeurIPS 2026

**Darius Muglich**, Niklas Lauffer, Tin Dizdarevic, Jakob Foerster

Recovered latent human conventions from unlabeled demonstrations to construct diverse evaluation partners, showing that partner coverage can change agent rankings.

**Expected Return Symmetries** ICLR 2025

**Darius Muglich**, Johannes Forkel, Elise van der Pol, Jakob Foerster

Learned expected-return-preserving symmetries beyond environment symmetries, enabling symmetry-based coordination without hand-specified transformations.

**Equivariant Networks for Zero-Shot Coordination** NeurIPS 2022

**Darius Muglich**, Christian Schroeder de Witt, Elise van der Pol, Shimon Whiteson, Jakob Foerster

Developed equivariant policy architectures and a test-time symmetrization operator to improve zero-shot coordination without retraining existing policies.

**Generalized Beliefs for Cooperative AI** ICML 2022

**Darius Muglich**, Luisa Zintgraf, Christian Schroeder de Witt, Shimon Whiteson, Jakob Foerster

Developed autoregressive transformer belief models that generalize to unfamiliar partner conventions and improve ad-hoc teamplay through belief-based search.

## Research and industry experience

**UC Berkeley · Center for Human-Compatible AI** Jun–Sep 2025

Visiting Researcher

Learned latent representations of human strategies from unlabeled demonstrations, enabling the construction of diverse AI partners for evaluating human–AI coordination.

**Bazaarvoice** Aug 2022–Sep 2026

Data Scientist V · Full-time initially; part-time during the doctorate

Primary engineer for a bandit-based campaign optimization method, delivering an estimated 40% lift in incremental revenue over an analyst-designed heuristic in a multi-month company A/B test.

## Selected awards

**Toshiba Research and EPSRC CDT Studentship** · Fully funded doctoral studentship

**NSERC Undergraduate Student Research Award** · Three awards

**Trek Excellence Scholarship** · Top 5% in year, faculty and school at UBC